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Completed	Friday, 16 October 2026, 5:07 PM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Inserting data into a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It complicates the selection of items
- c. It makes the data unsorted
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,80,90,150,200,300,400,470,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data in a single pass
- c. Sorting data sequentially
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Arranging data in a specific order
- c. Merging two datasets
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **7**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By comparing elements sequentially
- c. By using the bubble-up method
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **8**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Merging
- b. Rearranging
- c. Searching
- d. Sorting

The correct answer is: Sorting

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are split again into smaller sublists
- b. They are compared element by element
- c. They are combined to form a single sorted list
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **10**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. Each part is sorted independently
- c. Each part is searched for a specific element
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Searching
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it doesn‘t preserver the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys